

Significance of Biosphere Relative Contribution to Effective Complexity of the Universe

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Abstract

The observable Universe is overwhelmingly dominated by simple, large-scale physical regularities, yet it contains minuscule subsystems—most notably the biosphere of Earth—that exhibit extraordinarily high effective complexity. Drawing on Gell-Mann and Lloyd’s concept of effective complexity and the heuristic principle of relative contribution presented here, this essay argues that the biosphere, despite its negligible physical scale (approximately 10^{-39} of observable baryonic mass), makes a disproportionately large contribution to the describable regularities of the Universe as a whole. We provide order-of-magnitude estimates suggesting that biosphere-specific schemas may constitute approximately 50% of the minimal description of known cosmic regularities, despite the biosphere’s infinitesimal physical fraction. This asymmetry carries significant philosophical implications for ontology, epistemology, axiology, and our understanding of the place of life and mind in cosmic reality. We address observer-relativity concerns by considering how alternative intelligences might weight regularities differently, and refine the axiological argument to apply to any complexity-valuing observer.

Keywords: effective complexity, biosphere, information theory, minimum description length, hierarchical systems, philosophy of biology, cosmic significance

1 Introduction

The cosmological picture of the Universe is one of vast emptiness punctuated by hierarchical structures of matter: subatomic particles aggregate into atoms, stars form galaxies, galaxies cluster into superclusters, and the whole is woven into the cosmic web. On the largest observable scales the distribution is remarkably homogeneous and isotropic. Yet within this immense framework exists a tiny, thin film on one small planet—the biosphere—containing patterns of organization so dense and compressible that they stand out as qualitatively exceptional.

This essay explores the philosophical consequences of the following observation: when one seeks the shortest possible description of the regularities present in the observable Universe, the biosphere occupies a share far larger than its physical proportions would suggest. This disproportion is not accidental; it follows from a conceptual principle linking subsystem effective complexity to its relative contribution to the overall describable order of a hierarchical system.

A crucial clarification is necessary at the outset: the claim is not that the biosphere has high *physical* complexity (mass, energy, spatial extent) but high *effective* complexity—the information content of its compressible regularities. The biosphere’s mass is approximately 5.5×10^{14} kg (as carbon), while the observable Universe contains roughly 1.5×10^{53} kg of baryonic matter—a ratio of approximately 3.67×10^{-39} , or roughly 10^{-39} . Yet, as we shall argue, the biosphere’s contribution to the *describable regularities* of the Universe is vastly larger than this fraction suggests. The key insight is that regularity density (bits of compressible pattern per unit mass or volume) trumps sheer physical scale when constructing optimized descriptions.

The argument proceeds in four main steps: first, a restatement of the relevant scientific framework; second, quantitative estimates of the biosphere’s relative contribution; third, consideration of failure modes, counterarguments, and observer-relativity; and fourth, the philosophical implications that follow.

2 Scientific Framework: Effective Complexity and Relative Contribution

2.1 Effective Complexity

Gell-Mann and Lloyd ([Gell-Mann & Lloyd, 1996, 2003](#)) introduced effective complexity as a measure that captures the information content of an entity’s regularities, distinguishing structured patterns from random or incidental features. The effective complexity of an entity x , denoted $C_{\text{eff}}(x)$, is defined as the length of the shortest description of its regularities (schemas or patterns), after distinguishing them from incidental or random features. Total information decomposes as:

$$\Sigma(x) \approx C_{\text{eff}}(x) + I_{\text{random}} \quad (1)$$

where the first term captures compressible regularities and the second the incompressible residue (random or incidental information). The optimization criterion—minimize total information first, then effective complexity among equivalent descriptions—implies that salient regularities are preferentially retained in compact system descriptions. Ay, Müller, and Szkola ([Ay et al., 2010](#)) provided a rigorous formal definition and proved key properties, including the relation between effective complexity and logical depth—showing that strings with high effective complexity must have large computational depth. This connection to logical depth ([Bennett, 1988](#)) is significant: biosphere regularities are not merely complex but computationally deep, reflecting billions of years of evolutionary history that cannot be shortcut.

2.2 Hierarchical Structure of the Universe

A *hierarchical system* is one composed of interrelated subsystems, each of which may itself be hierarchical, down to some lowest level of elementary components ([Simon, 1962](#)).

A *modular system* is a hierarchical system in which subsystems exhibit strong internal coupling and weak external coupling—Simon’s “nearly decomposable” architecture.

The observable Universe exemplifies hierarchical organization across vast scales: quarks compose hadrons, which compose atomic nuclei; atoms form molecules; molecules aggregate into dust, planetesimals, planets, and stars; stars cluster into galaxies; galaxies form groups, clusters, and superclusters; and these are woven into the cosmic web of filaments and voids. At each scale, new regularities emerge from the interactions of components at lower scales. Recent network analyses have validated and extended Simon’s framework, demonstrating hierarchical modularity in systems ranging from metabolic networks (Ravasz et al., 2002) to human brain functional networks (Meunier et al., 2010).

2.3 The Heuristic Principle of Relative Contribution

By *relative contribution* of a subsystem to the system’s effective complexity, we mean the degree to which the subsystem’s regularities reduce the optimal description length of the whole when reused as building blocks or macros. This can be operationalized as: (a) the mutual information between the subsystem’s compressed description and the system-level description; (b) the reduction in total description length achieved by referencing the subsystem’s regularities rather than encoding them de novo; or (c) in toy models, the difference in minimum description length (MDL) with and without access to subsystem schemas.

The core insight can be stated as follows:

The Effective Complexity Principle: *High effective complexity in a subsystem tends to yield a significant relative contribution to the system’s overall describable regularities, as its rich internal patterns supply substantial compressible information for an optimized description of the whole system.*

The rationale proceeds as follows. An observer seeking to minimize description length will search for reusable patterns—regularities that compress multiple aspects of the system. Subsystems with high C_{eff} contain, by definition, many non-random, compressible regularities. When these regularities are salient at higher scales, the optimization process naturally incorporates them as macros or building blocks, reducing the description length required for the composite system. The subsystem thereby becomes a key explanatory element in the system-level schema.

This principle rests on several assumptions: (A1) the system admits a hierarchical or modular decomposition into identifiable subsystems; (A2) subsystems possess internal regularities that can be characterized by effective complexity; (A3) an observer constructs an optimized description of the whole system by minimizing total information; and (A4) subsystem regularities are at least partially non-redundant with respect to other subsystems and to emergent system-level patterns.

3 Quantifying the Disproportion: A Toy MDL Model

To move beyond intuition, we construct a simplified MDL comparison between biosphere-inclusive and biosphere-excluded descriptions of the Universe’s regularities. This is necessarily schematic—a full treatment would require consensus on the optimal encoding of physical and biological laws—but provides order-of-magnitude estimates.

3.1 Cosmological Schema (Biosphere-Excluded)

The minimal description of the non-biological Universe requires approximately:

- Standard Model of particle physics: ~ 25 free parameters $\times \sim 64$ bits each $\approx 1,600$ bits
- General relativity: field equations + cosmological constant ≈ 500 bits
- Λ CDM cosmological model: 6 parameters + initial conditions ≈ 800 bits
- Thermodynamics and statistical mechanics: ~ 300 bits
- Chemistry (periodic table, bonding rules): $\sim 1,000$ bits
- Astrophysical regularities (stellar evolution, nucleosynthesis): $\sim 1,500$ bits

Total cosmological schema: $\sim 5,700$ bits

This schema suffices to describe the regularities of approximately 10^{80} particles across 93 billion light-years—an extraordinary compression ratio.

3.2 Biosphere Schema (Additional Regularities)

The biosphere introduces regularities not derivable from cosmological schemas:

- Genetic code (64 codons $\rightarrow$ 20 amino acids + stops): ~ 300 bits
- Central dogma + molecular machinery (transcription, translation, replication): ~ 800 bits
- Mechanisms of hereditary variation and differential reproduction: ~ 200 bits
- Core metabolic pathways (glycolysis, Krebs cycle, electron transport): $\sim 1,200$ bits
- Cell theory and organelle organization: ~ 600 bits
- Developmental biology principles (morphogen gradients, Hox genes): ~ 800 bits
- Ecological principles (trophic levels, niche theory, island biogeography): ~ 700 bits
- Biogeochemical cycles: ~ 400 bits
- Neuroscience fundamentals (action potentials, synaptic transmission): ~ 500 bits
- Human-specific regularities (language universals, cultural evolution): $\sim 1,000$ bits

Total biosphere schema: $\sim 6,500$ bits

A note on terminology is warranted here. The phrase “mechanisms of hereditary variation and differential reproduction” deliberately avoids the term “Darwinian algorithm,” which implies a cumulative and creative role for natural selection that some researchers consider overstated ([Espeset, 2024](#); [von Sydow, 2012](#)). Natural selection undoubtedly occurs—differential survival and reproduction based on heritable traits is an observable phenomenon—but whether it constitutes a creative, cumulative algorithm capable of generating complex adaptations *de novo* remains contested. For the purposes of this analysis, we need only acknowledge that hereditary variation and differential reproduction are compressible regularities of the biosphere, without committing to any particular interpretation of their creative power.

3.3 Analysis

The combined schema totals approximately 12,200 bits, of which the biosphere contributes roughly 53%. This is a striking disproportion: a subsystem constituting 10^{-39} of cosmic mass contributes over half of the minimal description of known regularities.

Several important caveats apply. First, these estimates are illustrative order-of-magnitude figures; different encoding schemes could shift values by factors of 2–5, though the qualitative asymmetry would persist. The Standard Model parameter count (here ~ 25) aligns with standard physics estimates (19 core parameters, up to ~ 26 including neutrino oscillations), but the actual description length depends on encoding symmetries and field equations. Second, the cosmological schema compresses vastly more physical stuff than the biosphere schema, so bits-per-kilogram ratios differ enormously. Third, some biosphere regularities (biochemistry) partially derive from quantum chemistry and thermodynamics, reducing strict non-redundancy—though they are not *reducible* without loss of descriptive economy. Fourth, future unified theories might compress more overlap between physical and biological regularities than currently possible.

Nevertheless, the qualitative conclusion is robust: the biosphere’s contribution to describable regularities is grossly disproportionate to its physical scale. This conclusion finds independent support in assembly theory (Marshall et al., 2021), which measures the minimum number of joining operations required to construct complex objects. Biological molecules exhibit assembly indices far exceeding abiotic chemistry, reflecting precisely the kind of compressible, non-random regularity that effective complexity captures.

We can express this as a *regularity density* ratio. The cosmological schema describes $\sim 10^{53}$ kg with $\sim 5,700$ bits, yielding $\sim 10^{-50}$ bits/kg. The biosphere schema describes $\sim 5.5 \times 10^{14}$ kg with $\sim 6,500$ bits, yielding $\sim 10^{-11}$ bits/kg. The biosphere is roughly 10^{37} times more regularity-dense than the cosmos at large.

3.4 Compressible Regularities versus the Irreducibility of Individual Organisms

A crucial distinction must be drawn between the compressibility of *biological regularities* (the genetic code, metabolic pathways, developmental principles) and the irreducibility of *individual organisms* in their full particularity. The schemas enumerated above compress *types*—classes of phenomena that recur across the biosphere. But any individual organism, considered in its complete historical, relational, and phenomenological richness, resists such compression.

This point has been emphasized by several authors who argue that organisms are not “integrable” in the mathematical sense—they cannot be reduced to a point, as a physical object can be reduced to its center of mass or a geometric figure to its centroid (Longo et al., 2012; Montévil & Mossio, 2015). Each organism is a singular historical trajectory, shaped by contingent events, environmental interactions, and developmental noise that no schema can fully capture. The reduction of organisms to abstract quantities like “fitness” or “adaptation”—concepts that are notoriously difficult to define precisely—risks losing precisely what makes biological individuals distinctive (Espeset, 2024).

This distinction actually *strengthens* rather than weakens the argument of this essay. The biosphere contributes to the Universe’s effective complexity in two distinct ways: (1) through the compressible regularities that our MDL analysis captures, and (2) through the incompressible particularity of billions of individual organisms, each representing a

unique instantiation that adds to the total information content without adding to the schema length. The regularities provide the compression; the individuals provide the instantiation. Both are essential to the biosphere’s full contribution to cosmic order.

Our MDL estimates therefore represent a *lower bound* on the biosphere’s significance. They capture only the compressible regularities—the schemas—while the irreducible particularity of organisms constitutes an additional, perhaps even larger, contribution to the describable richness of the Universe. A complete description of reality must include both the laws and the instances, both the types and the tokens.

4 Failure Modes, Counterarguments, and Observer-Relativity

4.1 Standard Failure Modes

The heuristic principle may fail or be attenuated under certain conditions:

(F1) Emergent dominance: When relational or emergent regularities at the system level overshadow subsystem patterns. In the Universe case, cosmological regularities (dark energy, large-scale structure) are genuinely emergent and non-reducible to subsystem patterns. However, they do not *displace* biosphere regularities from the optimal description—they coexist as distinct schema components.

(F2) Redundancy: When multiple subsystems share similar regularities. If life exists elsewhere in the Universe with similar biochemistry, the biosphere’s contribution would be partially redundant. Current evidence—and arguments such as the Rare Earth hypothesis (Ward & Brownlee, 2000)—suggests Earth’s biosphere may be unique in the observable Universe, or at least extremely rare, maximizing its non-redundancy. Even if microbial life proves common, complex multicellular biospheres with the full hierarchy of regularities described here may remain singular.

(F3) Isolation: When a subsystem’s regularities are not referenced at higher scales. The biosphere is deeply integrated into Earth’s geochemistry (oxygen atmosphere, carbon cycle) and increasingly into global technological systems, ensuring strong coupling rather than isolation.

(F4) Observer dependence: Different descriptive frameworks may identify different regularities. This is the most serious concern and warrants extended treatment.

4.2 The Observer-Relativity Challenge: Alien Perspectives

Effective complexity is observer-relative: regularities are patterns that an observer identifies as compressible given their descriptive language and prior knowledge. Would non-human intelligences assign the same weight to biosphere regularities?

Consider several hypothetical observers:

Quantum-focused intelligence: An observer primarily interested in quantum phenomena might prioritize regularities in quantum field theory, quantum gravity candidates, or quantum information processing. Such an observer might find the biosphere’s classical-scale regularities less salient. However, even this observer would need to account for the biosphere if constructing a *complete* description of the Universe—the biosphere’s patterns exist whether or not they are prioritized.

Plasma-based intelligence: A hypothetical intelligence evolved in stellar plasma might have rich schemas for magnetohydrodynamic regularities that we barely perceive.

Such an observer might contribute additional schema components to the Universe description, but this would *add to* rather than displace biosphere regularities.

Pure mathematician: An abstract intelligence interested only in mathematical structure might discount all physical regularities equally—both cosmological and biological. This would not differentially affect the biosphere’s *relative* contribution among physical regularities.

The key insight is that observer-relativity affects *what counts as a regularity* but does not permit arbitrarily discounting regularities that demonstrably exist. Any observer capable of constructing Universe-descriptions must either (a) include biosphere regularities, thereby acknowledging their contribution, or (b) exclude them, thereby producing an incomplete description. The biosphere’s patterns are not projections of human interest but objective features of reality that any adequate description must encode.

4.3 The Anthropic Selection Objection

A deeper objection holds that we *inevitably* emphasize biosphere regularities because we are products of the biosphere—an anthropic selection effect. This is true but does not undermine the argument. The anthropic situation explains *why we notice* the disproportion, not whether the disproportion exists. An alien observer would face an analogous situation: whatever regularities produced them would be salient to them, yet those regularities would objectively exist.

Moreover, the anthropic objection proves too much: it would equally undermine any claim about the Universe, since all our knowledge is conditioned by our existence. The appropriate response is not skepticism but acknowledgment that all descriptions are constructed from particular perspectives—and that some descriptions are nonetheless more complete than others.

5 The Biosphere as a Disproportionately Significant Subsystem

Having established the conceptual framework and addressed major objections, we can now characterize the biosphere’s significance more precisely.

The biosphere is physically negligible. Its total biomass is roughly 550 gigatons of carbon (5.5×10^{14} kg), confined to a layer rarely thicker than a few kilometres on a planet whose mass (6×10^{24} kg) is already insignificant compared with the Sun (2×10^{30} kg), let alone the Milky Way (1.5×10^{42} kg) or the observable Universe (1.5×10^{53} kg baryonic). The biosphere constitutes approximately 10^{-39} of observable baryonic mass.

Yet the regularities it harbours are among the most elaborate and compressible known to science. A partial list of the biosphere’s major compressible regularities includes: the near-universal genetic code and its associated molecular machinery; the mechanisms of replication, variation, and differential reproduction; the hierarchical organization of metabolism, cellular structure, multicellularity, and ecosystems; global biogeochemical cycles (carbon, nitrogen, oxygen, phosphorus); emergent patterns such as trophic cascades, symbiosis, niche construction, and co-evolution; and in the human domain, symbolic language, cumulative culture, science, technology, and reflective consciousness.

These regularities are not merely complicated; they are highly compressible. A relatively compact description—consisting of the laws of biochemistry, ecological principles,

and a few boundary conditions—can account for an astonishing diversity of phenomena across scales from nanometres to the planetary surface.

In terms of the heuristic principle, the biosphere is an extreme case: high effective complexity combined with strong non-redundancy and deep integration into planetary boundary conditions yields an exceptionally large relative contribution to the overall regularities of the observable Universe. The biosphere’s regularities satisfy all the conditions for maximal contribution: they are dense (high C_{eff}), unique (non-redundant with cosmological regularities), and salient (deeply coupled to planetary and now global processes).

6 Philosophical Implications

6.1 Ontological: Life as a Fundamental Mode of Cosmic Organization

The disproportionate representational weight of the biosphere challenges the ontological privileging of “fundamental” physics over emergent biological order. If the shortest true description of reality must devote substantial space to living systems, then life is not a marginal or accidental phenomenon but one of the principal ways in which the Universe expresses order.

This does not imply vitalism or dualism; it is fully compatible with physicalism. Yet it undermines the reductionist intuition that “higher” levels are ontologically derivative or less real. The biosphere’s regularities are emergent, but they are not eliminable without catastrophic loss of descriptive economy. In Anderson’s famous phrase, “more is different” ([Anderson, 1972](#))—and in this case, “more” (in the form of biological organization) is dramatically different.

6.2 Epistemological: The Inevitable Bias of the Embedded Observer

Effective complexity is observer-relative: the choice of regularities depends on the language, prior knowledge, and interests of the describer. Because every known observer capable of constructing such descriptions is a product of the biosphere, there exists an unavoidable anthropic selection effect. We are predisposed to notice and emphasize the regularities that produced us.

This is not mere bias to be corrected; it is a structural feature of inquiry. Any description of the Universe that aspires to minimality must still be constructed from within the biosphere. Thus the disproportionate significance we assign to life may be both inevitable and legitimate: it reflects the fact that the most sophisticated measuring instrument the Universe has yet produced is itself a biological phenomenon.

6.3 Axiological: The Value of Preserving Complexity Generators

If the biosphere contributes so disproportionately to the describable order of the Universe, then its persistence and flourishing acquire more than parochial significance. The destruction of large fractions of biological diversity would erase compressible regularities that are not replicated elsewhere in the observable Universe.

To avoid anthropocentrism, we can frame this axiologically as follows: *for any intelligence that values describable order (regardless of its physical substrate or evolutionary origin), the preservation of high- C_{eff} systems increases the total regularities accessible to description.* This formulation does not assume human values but applies to any observer that finds compressed descriptions valuable—a minimal rationality criterion.

This confers a weak but genuine cosmic axiology: actions that safeguard or expand the generative capacity of life-bearing systems increase the total describable order accessible to any future intelligence. The preservation of biodiversity and the responsible stewardship of Earth’s habitability become, in a limited but non-trivial sense, contributions to the intelligibility of the cosmos itself—not because humans value them, but because any complexity-valuing observer would.

6.4 Metaphysical Restraint

One might be tempted to read teleological significance into this disproportion—to see the Universe as “aiming” at the production of complex observers. Such readings exceed what the evidence warrants. The principle of relative contribution is descriptive, not prescriptive; it identifies a structural feature of optimized descriptions, not an intention behind cosmic evolution.

Nevertheless, the fact that physical processes have produced a subsystem capable of generating and sustaining such dense regularities remains one of the most striking features of reality. Whether these processes are entirely “blind” or whether evolutionary mechanisms themselves exhibit forms of directionality, plasticity, or self-organization that transcend simple randomness is a matter of ongoing debate (Espeset, 2015; von Sydow, 2012; Jablonka & Lamb, 2005). For the purposes of this argument, we need not adjudicate this question: whatever the nature of the processes involved, the *result*—a subsystem of extraordinary regularity density—invites wonder without requiring metaphysics.

7 Conclusion

The biosphere is physically insignificant yet descriptively dominant. The heuristic principle of relative contribution to effective complexity provides a conceptual mechanism for understanding this asymmetry: rich, compressible regularities in a tiny subsystem can dominate the minimal description of an immense hierarchical whole. Order-of-magnitude MDL estimates suggest the biosphere contributes roughly half of the known regularities of the Universe while constituting 10^{-39} of its baryonic mass—a regularity density ratio of approximately 10^{37} .

This insight does not solve the mystery of life’s place in the Universe, but it reframes the question. Rather than asking whether life is cosmically important despite its small scale, we should ask whether any description of the Universe that aspires to completeness can afford to marginalize the richest known source of order it contains.

In the end, the disproportionate significance of the biosphere is not a paradox to be resolved, but a fact to be contemplated—one that invites us to view the Universe not only as a physical expanse, but as an unfolding tapestry of increasing describable richness, within which life plays an unexpectedly central role.

Acknowledgments

I am grateful to David Espeset for insightful comments on an earlier draft of this manuscript. His critical observations regarding the status of the “Darwinian algorithm,” the question of whether evolutionary processes are truly “blind,” and the crucial distinction between compressible regularities and the irreducible particularity of individual organisms have substantially improved the philosophical precision of this essay. Responsibility for any remaining errors or contentious claims remains entirely my own.

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